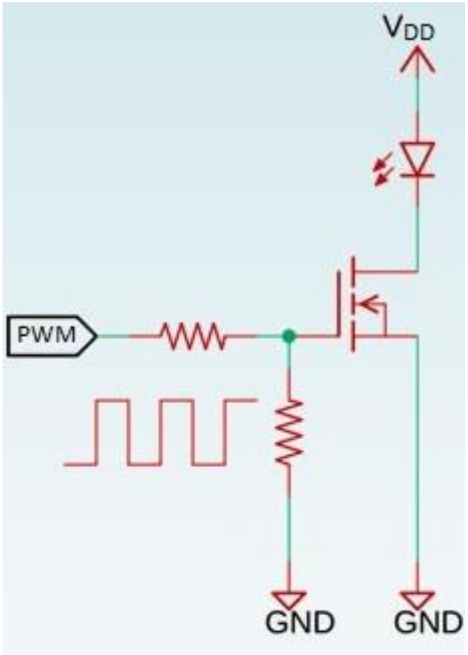
DC-DC CONVERTERS – BUCK CONVERTERS 2 (Design cont.)

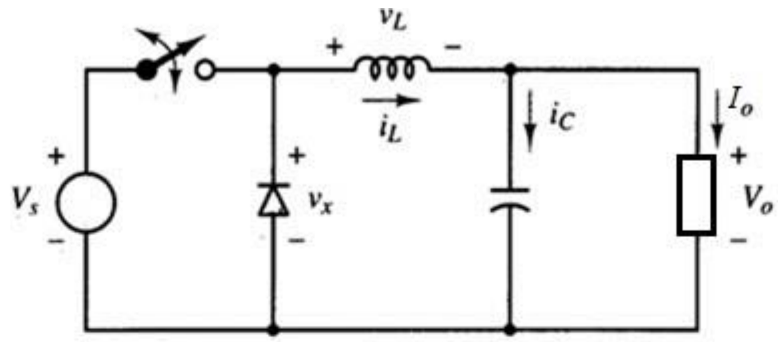
BUCK2-4 Switch (MOSFET) Circuit Placement

Very often (including for the Solar Car project), we utilise a MOSFET for the controlled switch. Now we need to consider the control conditions required to turn the MOSFET on and off. Let's start with some general considerations.

- V_{gs} needs to be large enough to turn MOSFET fully on (often ~10V)
- Gate-Source acts as a small capacitance load for input signal. So signal source needs to be able to source and sink a significant pulse current to turn MOSFET on and off quickly.



Consider again the basic Buck converter topology. The position of the controllable switch is a problem (for the project) if using a MOSFET. For the MOSFET to be switched fully on, the gate voltage needs to be about 10V higher than the source voltage. But when "on" the MOSFET source terminal voltage is nearly equal to the solar panel voltage (no voltage drop across the switch). We don't have the extra 10V above the panel voltage needed to turn the MOSFET on.



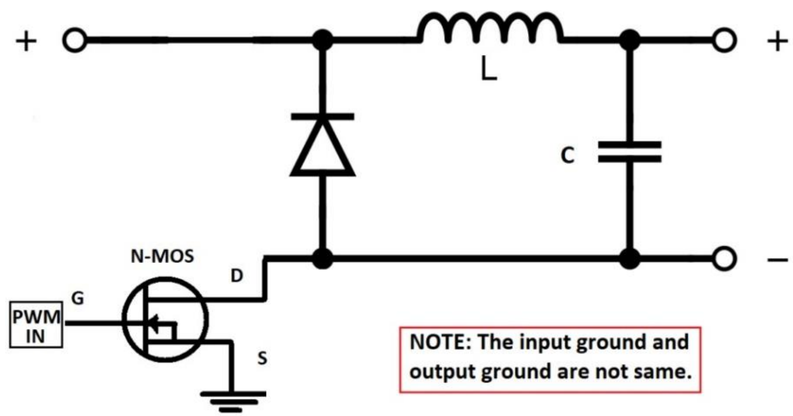
Q. How do we solve this problem?

A. We use what we know about the application,

and what we know about series-connected circuits.

Move the MOSFET from the "High-side"

to the "Low-side".



BUCK2-4.1 MOSFET Gate Drive (Solar Car Project)

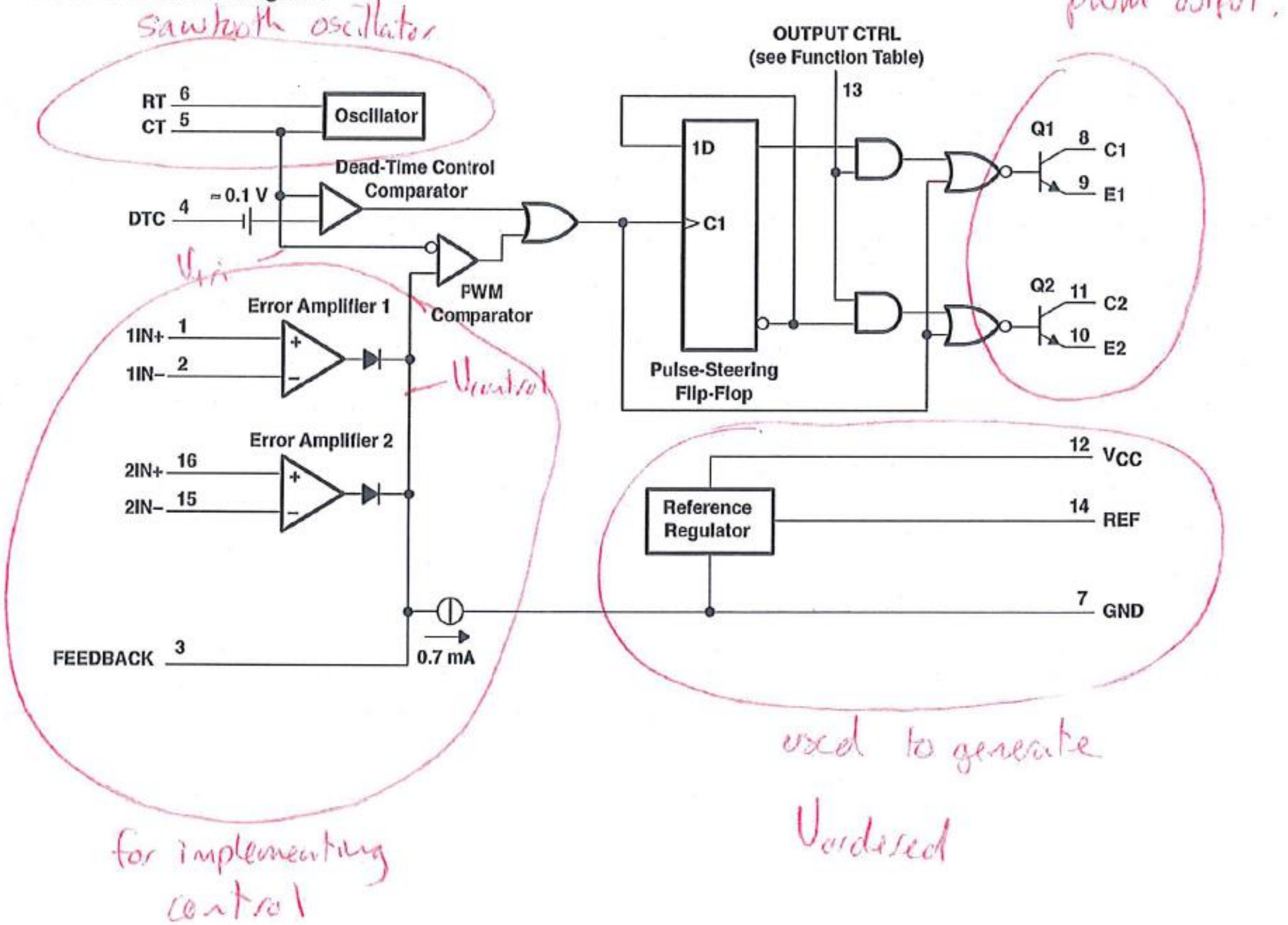
How are we going to make sure we turn the MOSFET on and off quickly?

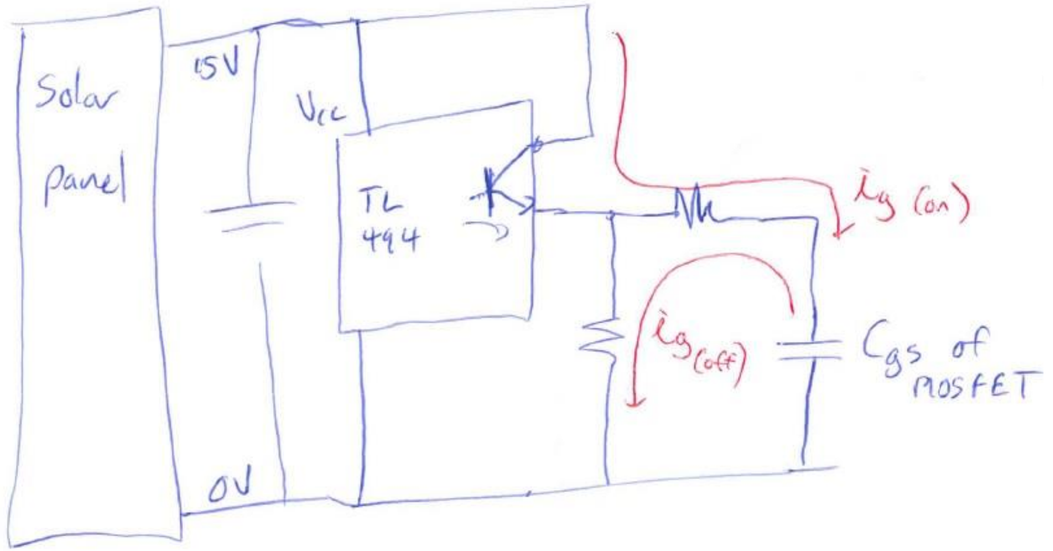
FUNCTION TABLE

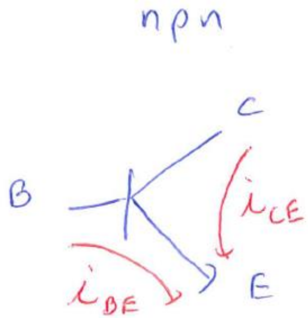
INPUT TO OUTPUT CTRL	OUTPUT FUNCTION
V_I = GND	Single-ended or parallel output
$V_I = V_{ref}$	Normal push-pull operation

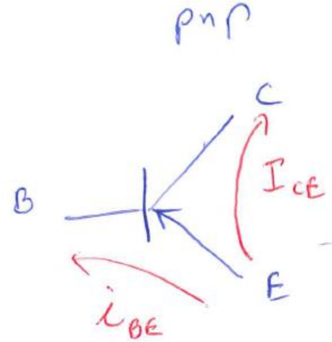
used to generate pwm output.

functional block diagram





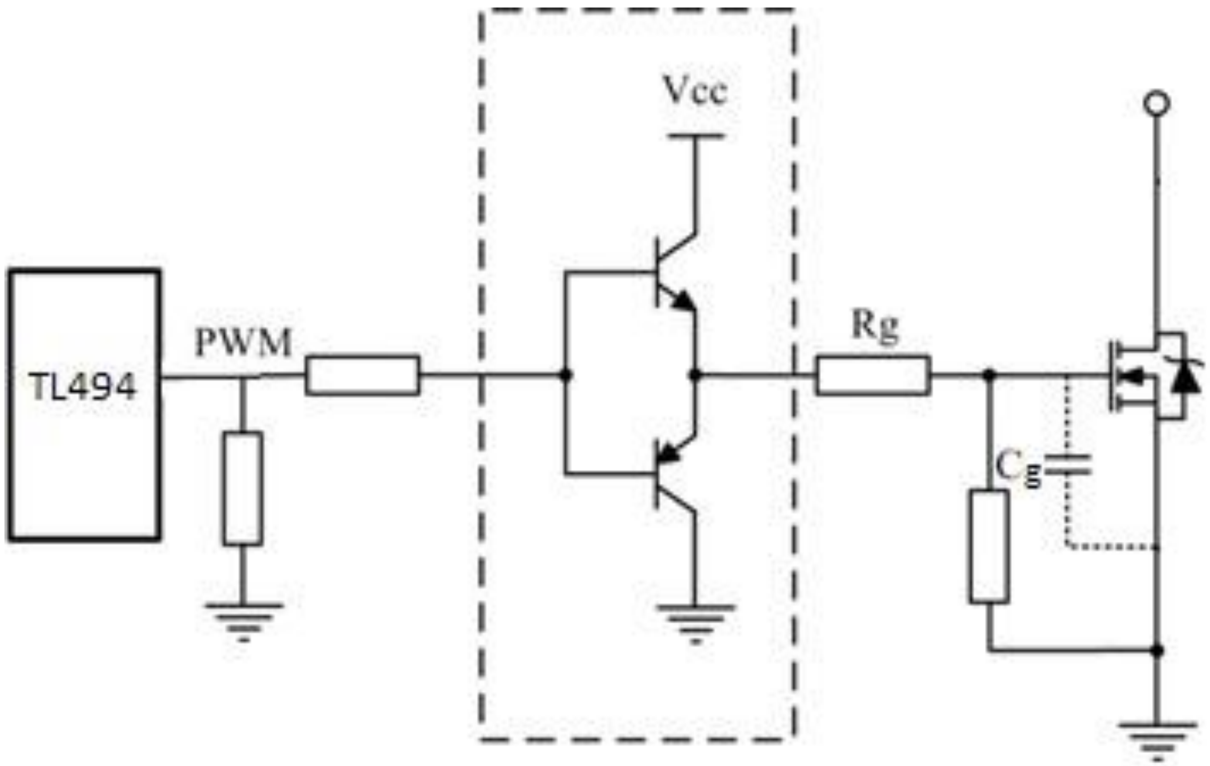




$$i_{CE} = \beta\, i_{BE}$$

$$10 < \beta < 100$$

One possible solution



DC-DC CONVERTERS – BOOST CONVERTERS

The ***boost converter*** uses essentially the same set of components as the buck converter, just arranged slightly differently as shown in Figure 1.1. Called 'boost' since $V_O > V_S$.

Remember that we are looking at all these DC-DC converters with the following **steady state** properties;

- The inductor current fluctuations are periodic $\{i_L(t+T) = i_L(t)\}$.
- The average inductor voltage is zero.
- The average capacitor current is zero.
- The power supplied by the source equals the power delivered to the load (plus any non-ideal losses).
- The filter capacitor is large enough to make any output voltage ripple mostly negligible.

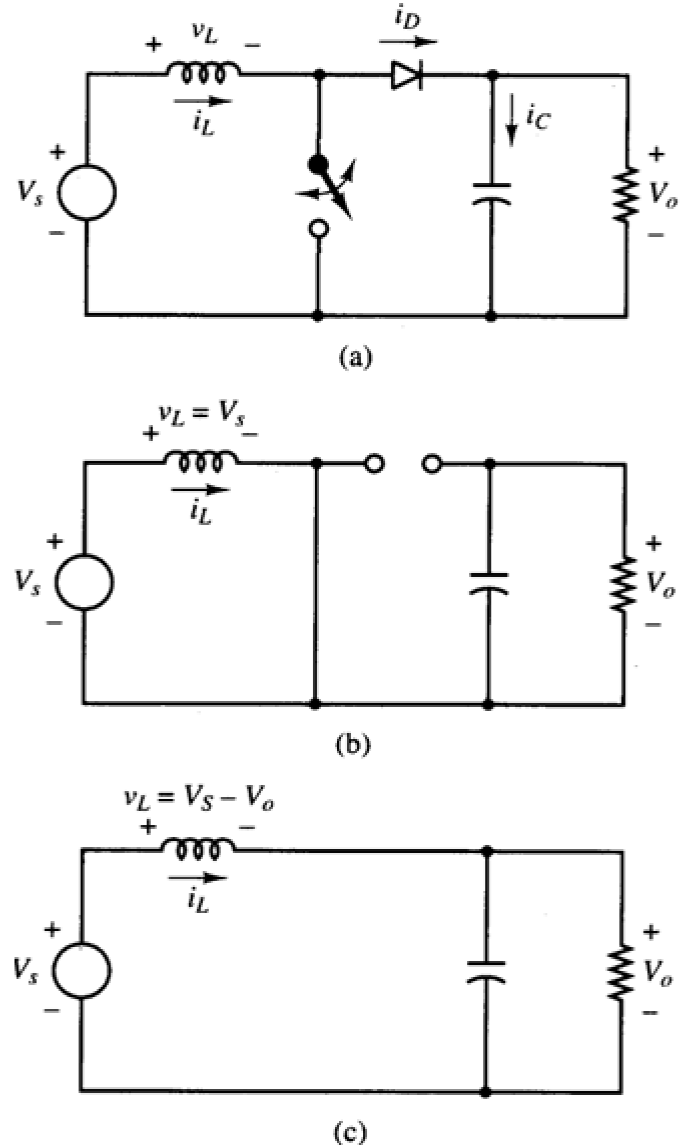
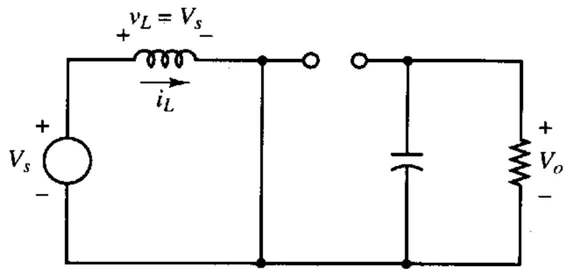


Figure 1.1 (a) Boost converter topology. (b) Switch closed. (c) Switch open.

Switch Closed



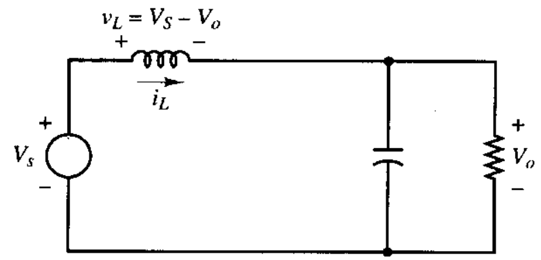
When the switch is closed, the diode is reverse biased which results in,

$$v_L = L\frac{di_L}{dt} = V_S \quad or \quad \frac{di_L}{dt} = \frac{V_S}{L}$$

Since V_S **and** L are constants, then the time derivative of the inductor current is a constant. That means, the inductor current must be **constantly ramping up** while the switch is closed, where

$$\Delta i_{L,closed} = \frac{V_S DT}{L}$$

Switch Open



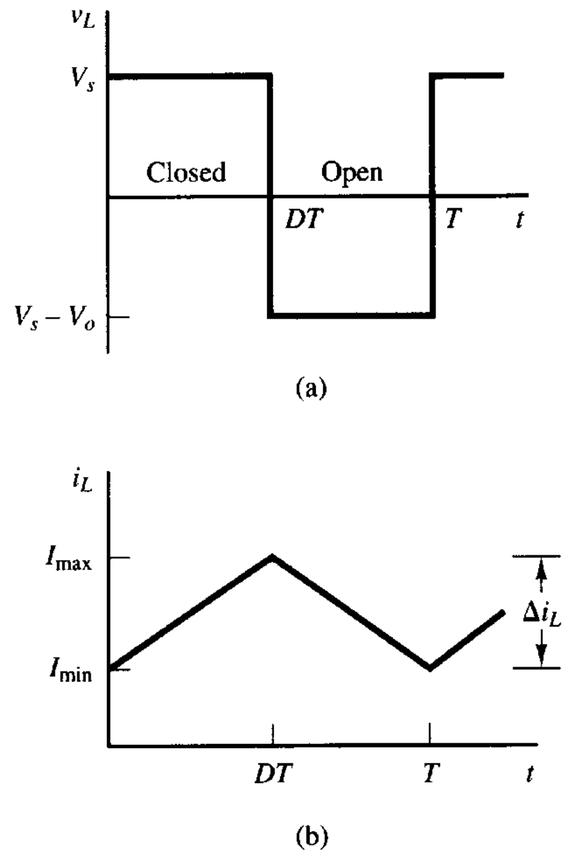
When the switch is open, the inductor voltage reverses to keep the current flowing in the same direction, which forward biases the diode. The inductor is effectively in series with the load so the inductor voltage adds to the source voltage to give the output voltage. So,

$$v_L = V_S - V_O = L\frac{di_L}{dt} \quad or \quad \frac{di_L}{dt} = \frac{V_S - V_O}{L}$$

Again the derivative of the current is a constant, so the current from the inductor is **constantly ramping down** while the switch is open, where

$$\Delta i_{L,open} = \frac{(V_S - V_O)(1-D)T}{L}$$

Since the average inductor voltage must equal zero (steady-state), then

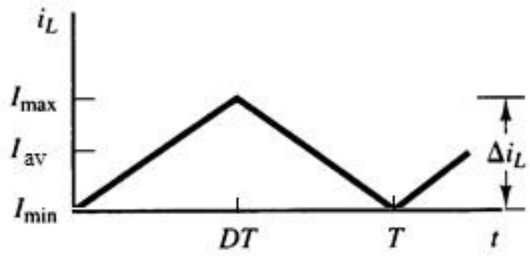


$$\boxed{V_O = \frac{V_S}{1-D}}$$

And since D lies somewhere between 0 to 1, the output voltage is always **greater** than or equal to the source voltage.

Continuous Current Design Consideration

To keep the inductor current continuous (continuous conduction), its minimum current must not drop to zero. So, since the inductor is always connected to the input source, the average input current (which is the same as the average inductor current) must be larger than half the change in inductor current.



In order to determine the average input current, the average output current can be used (assuming 100% converter efficiency), where,

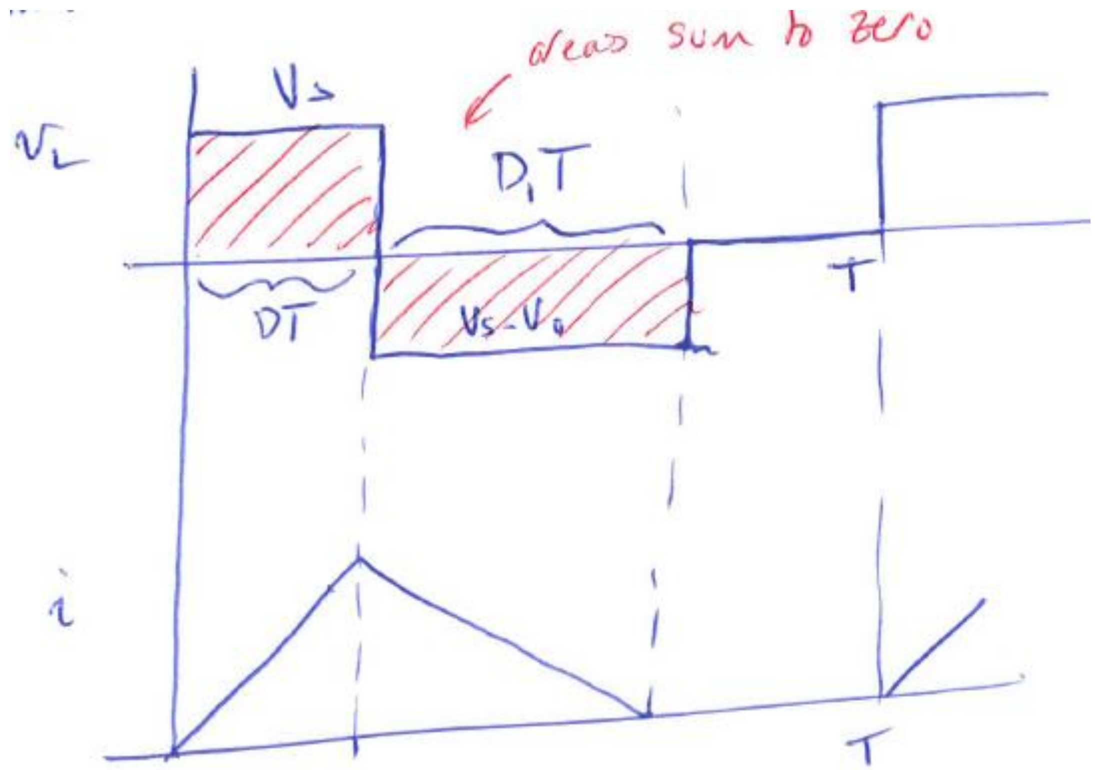
$V_s I_s = V_o I_o$ and ${}^{V_o}\!/_{V_s} = {}^{1}\!/_{1-D}$

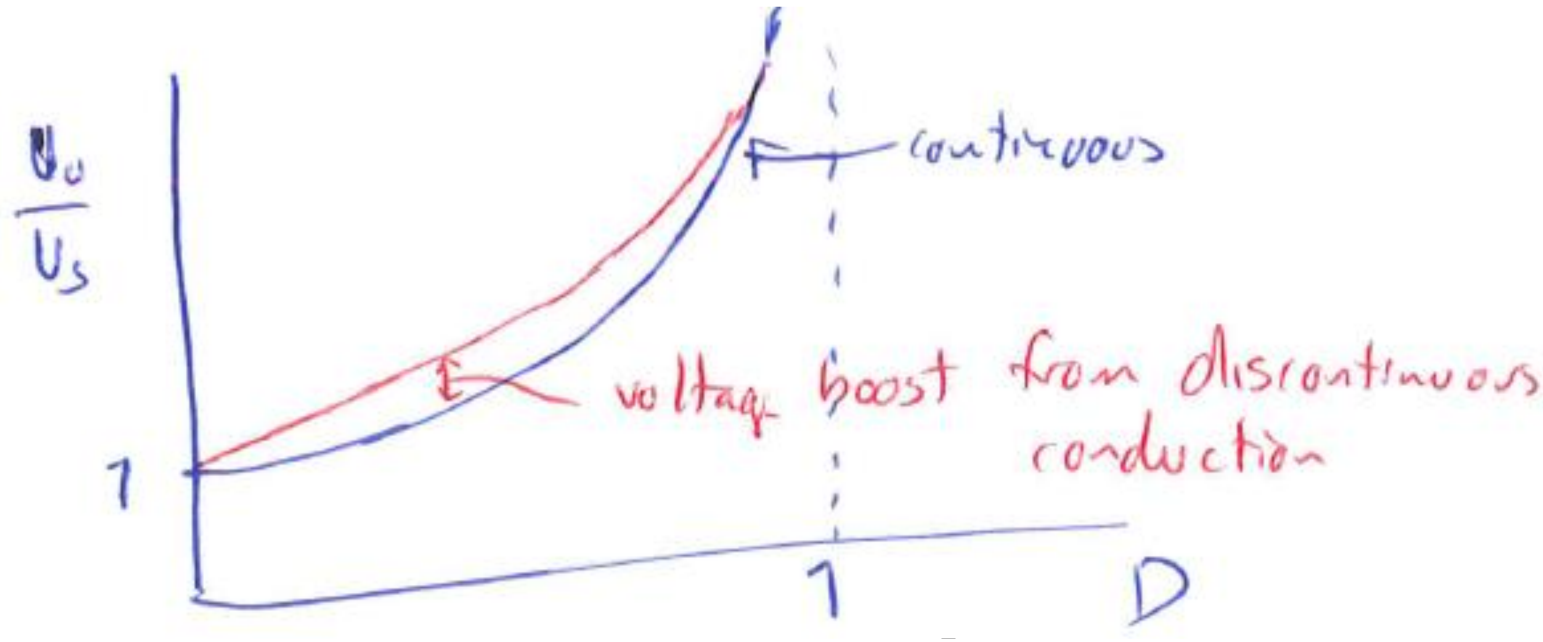
Hence,

$$L > \frac{V_s D(1-D)}{2 f_s I_o}$$

Discontinuous conduction

Want to avoid for all the same reasons as for the buck converter.

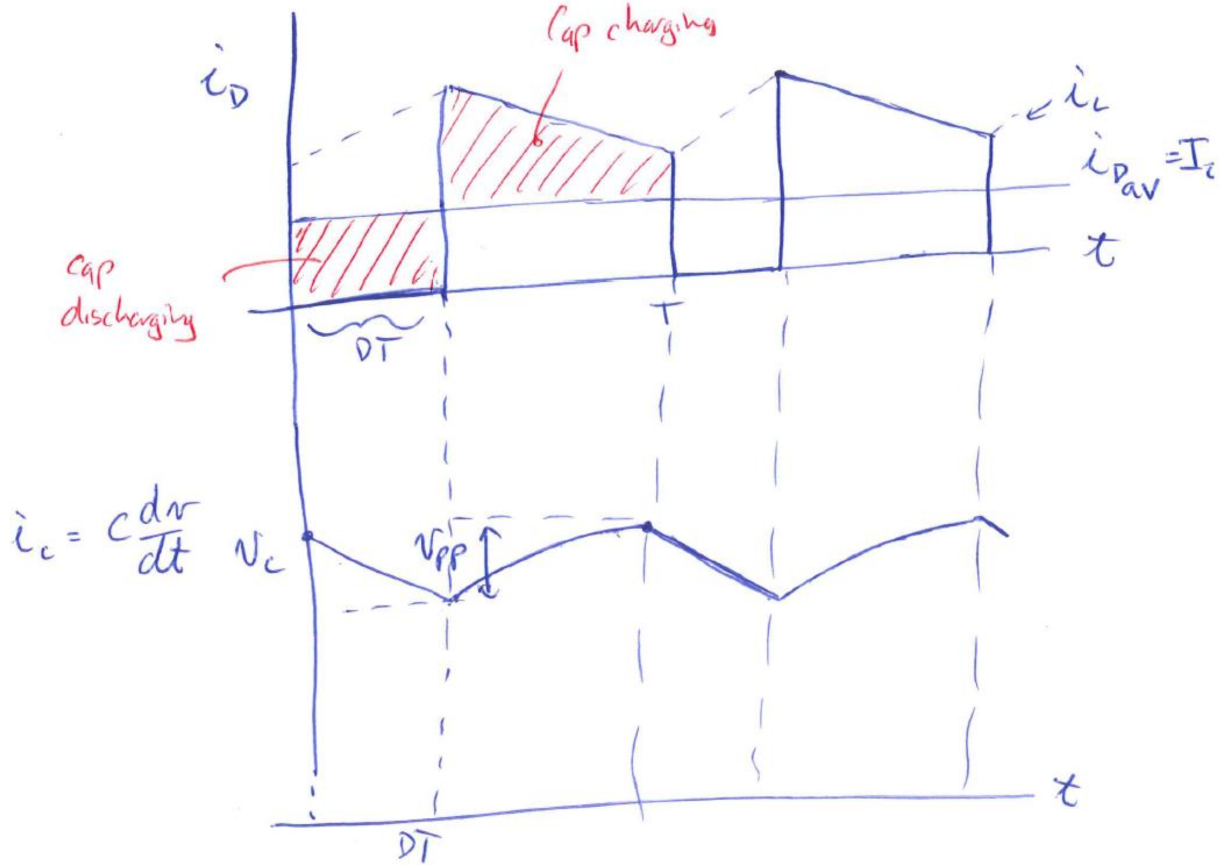




Output Voltage Ripple (Capacitor Sizing)

We now consider the more practical situation for the output voltage where there is some ripple, but is small enough not to have an appreciable effect on the previous analysis.

Again, if there is current flowing in the capacitor, then there must be an associated voltage variation. Consider diode current, load current, capacitor current, and capacitor voltage waveforms,



$$\Delta V_o = \frac{I_o D}{f_s C}$$